MSc Ecology and Data Science · Research Project BIOS0057

Which penguin is this?

Automatic identification of Humboldt penguins at ZSL London Zoo — and what an honest evaluation says it can really do.

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2,307
photographs

81
individuals

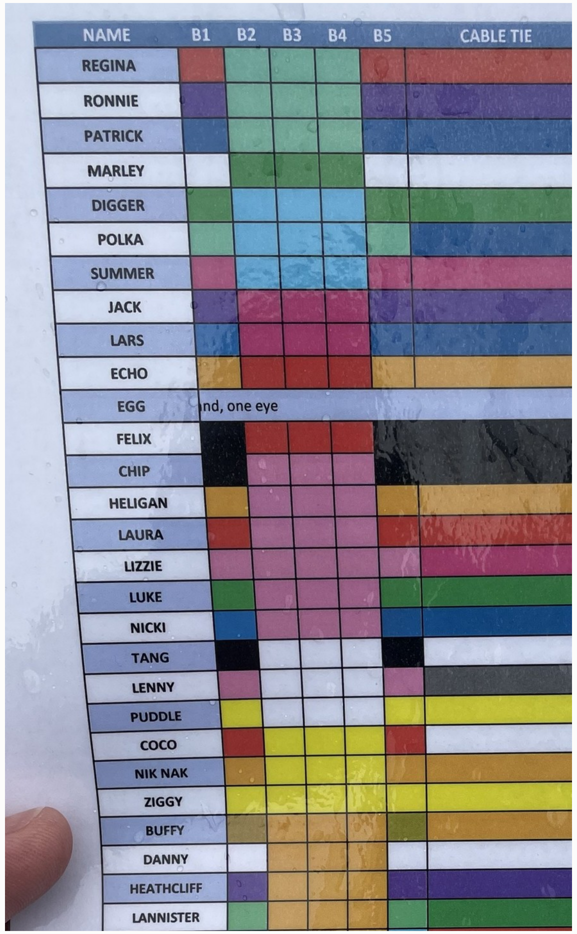
335
capture sessions

A penguin's name lives on a laminated sheet

ZSL London Zoo keeps 81 Humboldt penguins. A welfare note, a veterinary history or a breeding decision only means something once it is attached to the right bird.

- Coloured flipper bands are the current answer — and they are hidden by the bird's posture, by another penguin, or simply by the viewing angle.
- Over ten years, flipper-banded king penguins showed lower survival and breeding success than electronically tagged controls (Saraux et al., 2011). A marker is not automatically neutral.
- The birds already carry an identifier: the black spots on their white ventral plumage. African penguins use those same patterns to recognise each other (Baciadonna et al., 2024).

Ventral spot patterns: individual, stable, and already on the bird.



The colony's own band chart — the system a photograph would complement.

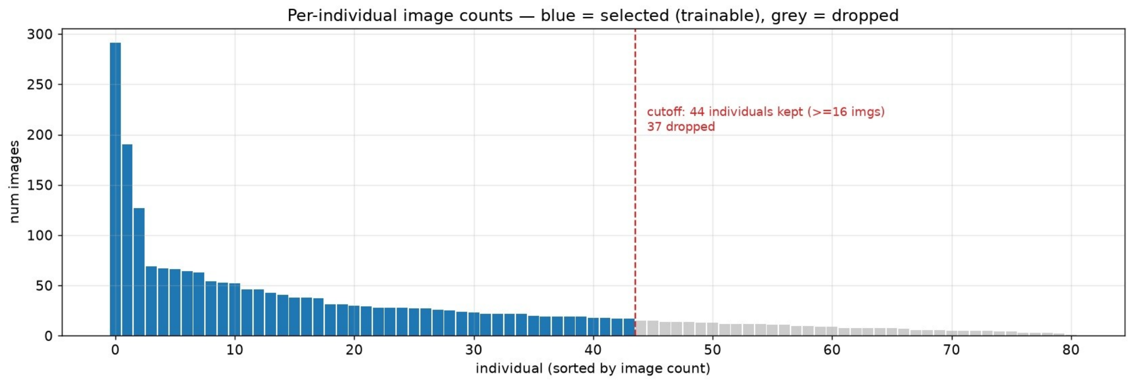
An archive, not an experiment

2,307
photographs

81
individuals

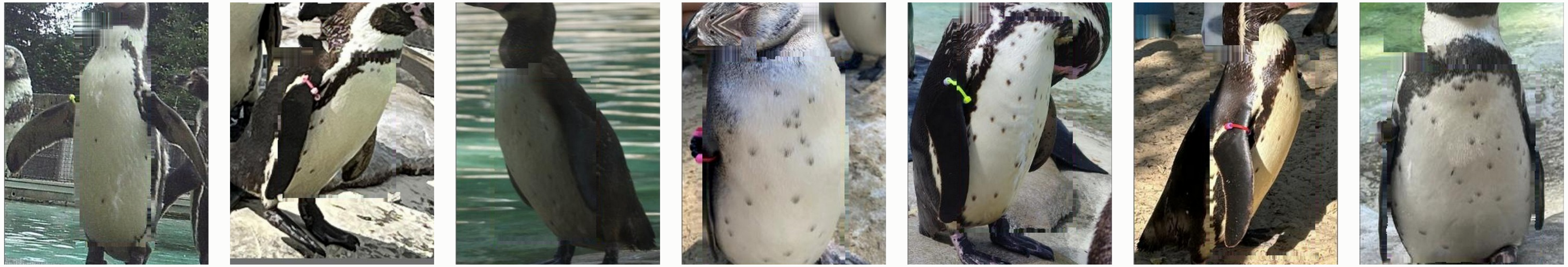
1 → 291
photographs per bird

335
capture sessions



Every colony member, sorted by photograph count — Medici holds 291.

Taken by volunteers during routine husbandry — no protocol, no fixed viewpoint, no schedule. Two inclusion thresholds leave the **35 individuals and 1,743 photographs** behind every number here.



Seven of the eighty-one, as photographed — different days, light, distance and backgrounds, and one bird facing away.

THE SYSTEM

A photograph goes in; a name — or an honest refusal — comes out

1. **Photograph** — frontal, full body, as taken
2. **Embedding** — ResNet18, 512-d, L2-normalised
3. **Gallery search** — FAISS, cosine over 81 prototypes
4. **Threshold 0.938** — accept, or decline to identify
5. **Decision** — identity + ranked alternatives

Why retrieval rather than an 81-way classifier

Enrolment, not retraining

A new bird is a stored vector. Smew and Skunk are searchable in the deployed gallery although no model was ever trained on them. A softmax head would need a new output layer and a full retrain for every arrival.

A principled way to refuse

Cosine similarity to a prototype carries a threshold. A classifier must spread probability across the names it knows, so it names something — whatever walks past the camera.

WHERE THIS PROJECT NEARLY ENDED

0.950

Test accuracy. ResNet18 over 44 individuals, random 70:15:15 split.

Frozen features plus FAISS retrieval scored 0.959. Both looked like a finished system.

It was not.

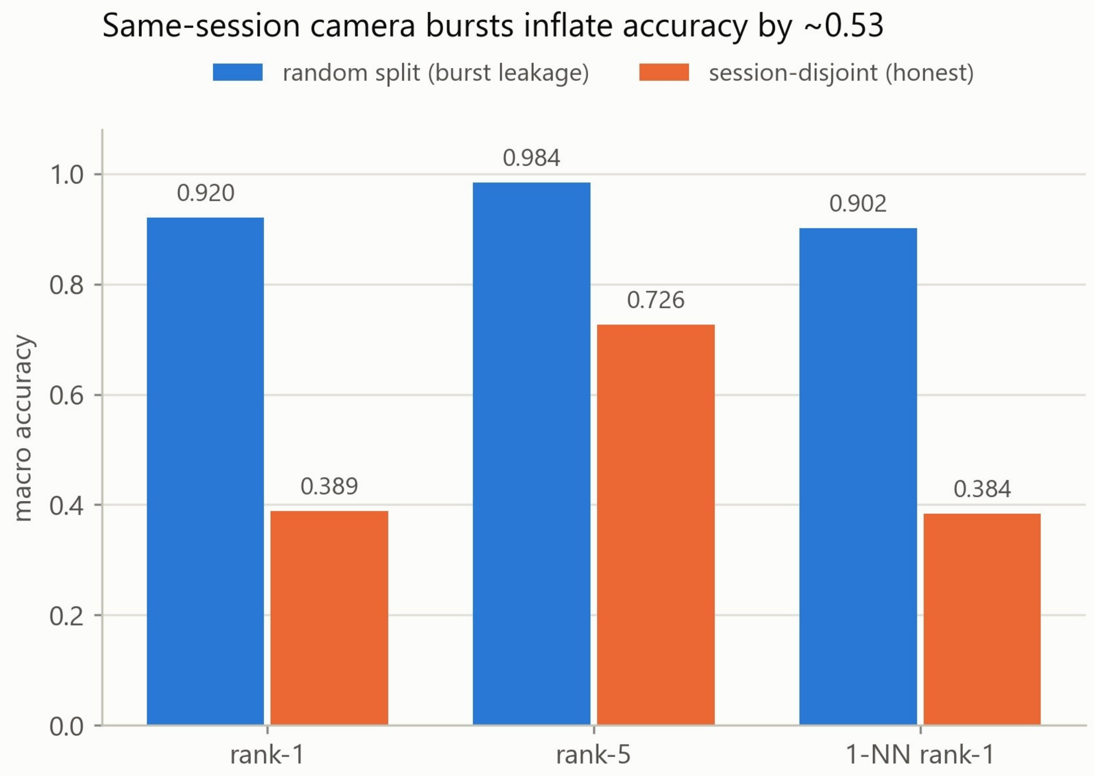
Then I looked at what was in the test set

The archive is full of bursts — DSC_2743 … 2749: one camera, one moment, seven frames. A random split scatters those frames across train and test.

- 97.8% of test photographs shared a capture session with a photograph in the gallery.
- 13.8% had an outright near-duplicate, by perceptual hash and pixel correlation.
- Of the EXIF-stamped photographs, ~29% were within one second of their nearest training image.

Same 35 birds. Same 1,743 photographs. Same per-bird partition sizes. Same training recipe. Only the splitting rule changed.

Absolute inflation: **0.531**



Macro accuracy: each individual weighted equally, whatever its photograph count.

An evaluation that cannot flatter itself

1 The unit is the encounter

A capture session is one photographic occasion, inferred from filename prefix and a frame-number gap above 50, cross-checked against EXIF. 335 sessions across the colony.

2 Sessions are dealt into folds, never photographs

Session-wise five-fold cross-validation. All 1,743 photographs are tested exactly once, each by a model that never saw its own session — against 261 under a single split.

3 Every bird counts once

Macro averaging throughout: Medici's 291 photographs weigh exactly as much as Beau's 19. Per-image accuracy silently adopts a prior over which penguin gets photographed.

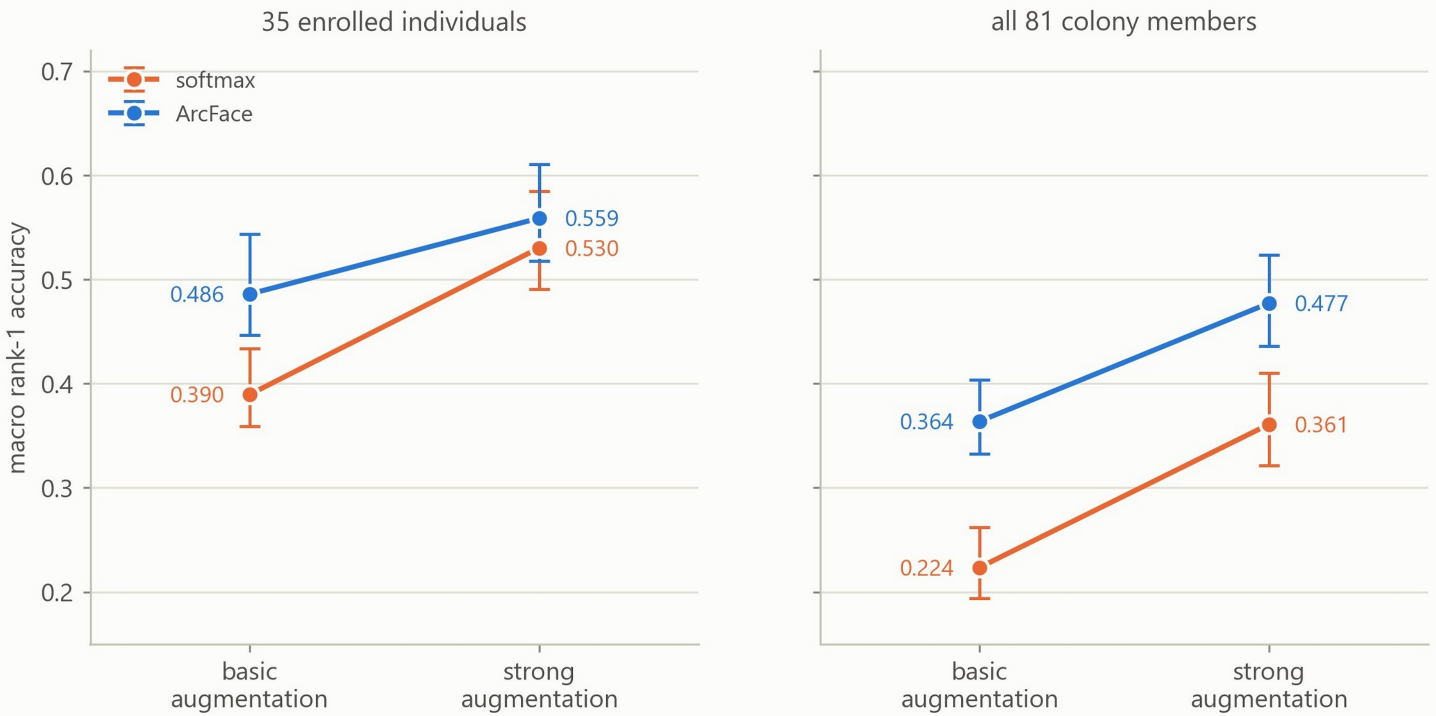
4 Uncertainty respects the bursts

95% intervals resample capture sessions, not photographs. Fixed 80 epochs, no validation split, no early stopping, no checkpoint selection — so nothing is tuned per configuration.

Pre-registered before any run and held identical for every configuration compared — so the comparisons below measure the change, not the tuning.

Which change actually did the work

- Strong augmentation alone
  +0.141 [+0.105, +0.179]
- ArcFace alone
  +0.096 [+0.061, +0.139]
- ArcFace given strong aug.
  +0.029 [−0.005, +0.060] n.s.



Best configuration

0.559

ArcFace + strong augmentation, macro rank-1 over 35 individuals

Augmentation is the larger single lever — and the two are sub-additive, interaction −0.068. But ArcFace's marginal value grows with the gallery: invisible at 35 candidates, +0.116 against all 81. Metric learning shapes the embedding geometry, not the top-1 call.

0.559 is the friendliest number in this talk

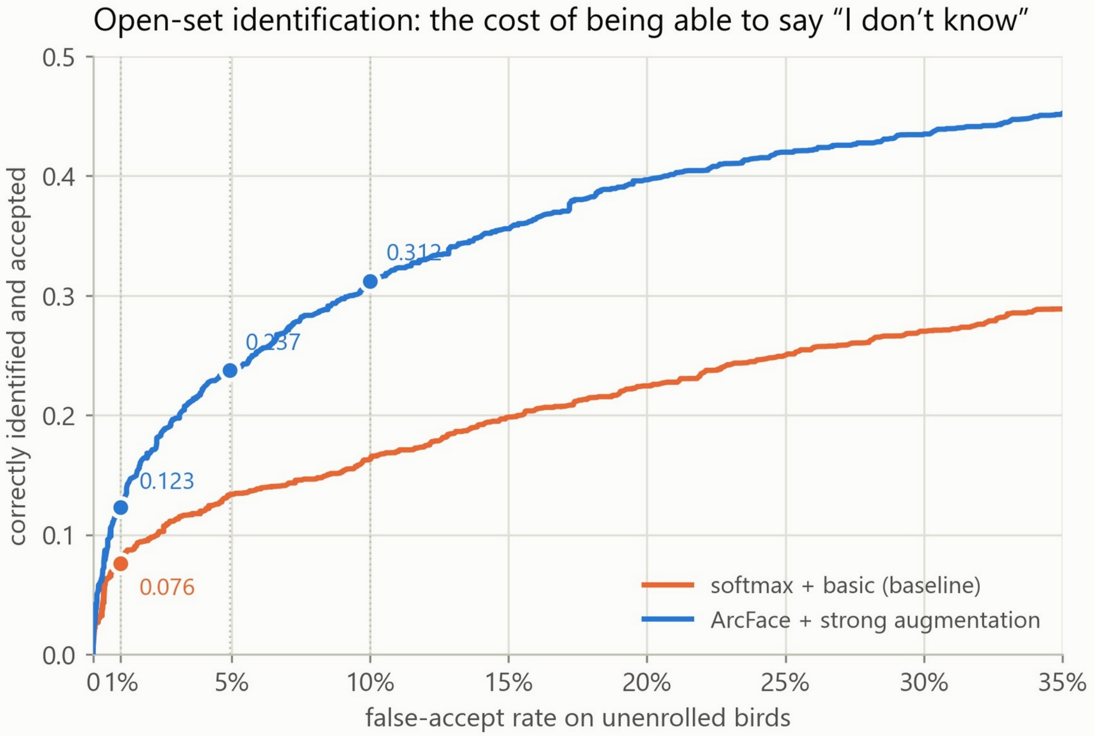
- **How big is the line-up?**

  0.559 assumes 35 candidates. Add the other 46 colony members as distractors — no retraining, same queries — and it falls to 0.477. ArcFace degrades least: −0.082, against the baseline's −0.166.

- **What if the bird is not enrolled?**

  564 photographs of 46 birds no model ever trained on. Required to refuse those strangers 99% of the time, the system still correctly names only 0.123 of enrolled birds.

Simulating strangers instead claimed 0.230 — an 87% overestimate. The earlier 0.80 threshold, tuned that way, actually accepts 29.7% of real strangers.



Strangers are real: 46 colony members the cross-validated models never trained on.

Nicki and Gonzo

Nicki

53 photographs · 3 sessions

0.189

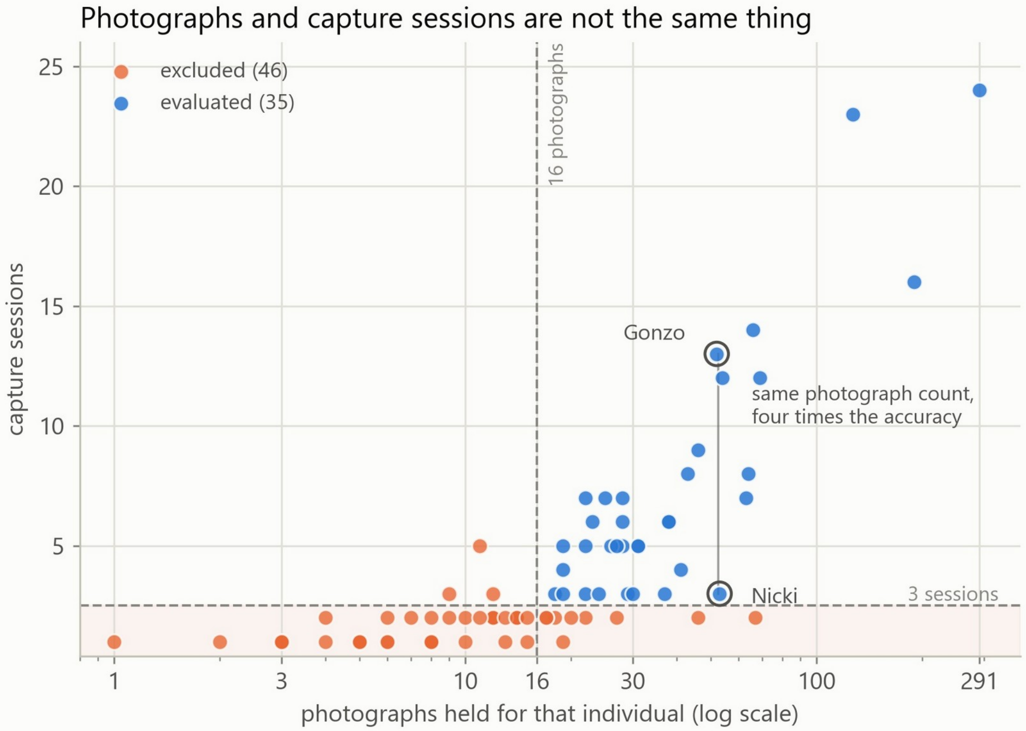
Gonzo

52 photographs · 13 sessions

0.827

Almost the same number of files. Four times the accuracy.

- Grouped by session count: 3 sessions (8 birds) 0.186 · 4–6 (13 birds) 0.618 · 7+ (14 birds) 0.717.
- Controlling for photograph count, sessions still predict accuracy: partial ρ = 0.518, p = 0.002. Controlling for sessions, photographs do not: ρ = 0.224, p = 0.203.
- Metric learning lifts every bucket but cannot rescue the three-session birds — from 0.081 only to 0.186.



One point per colony member. Photograph count runs across; capture sessions run up.

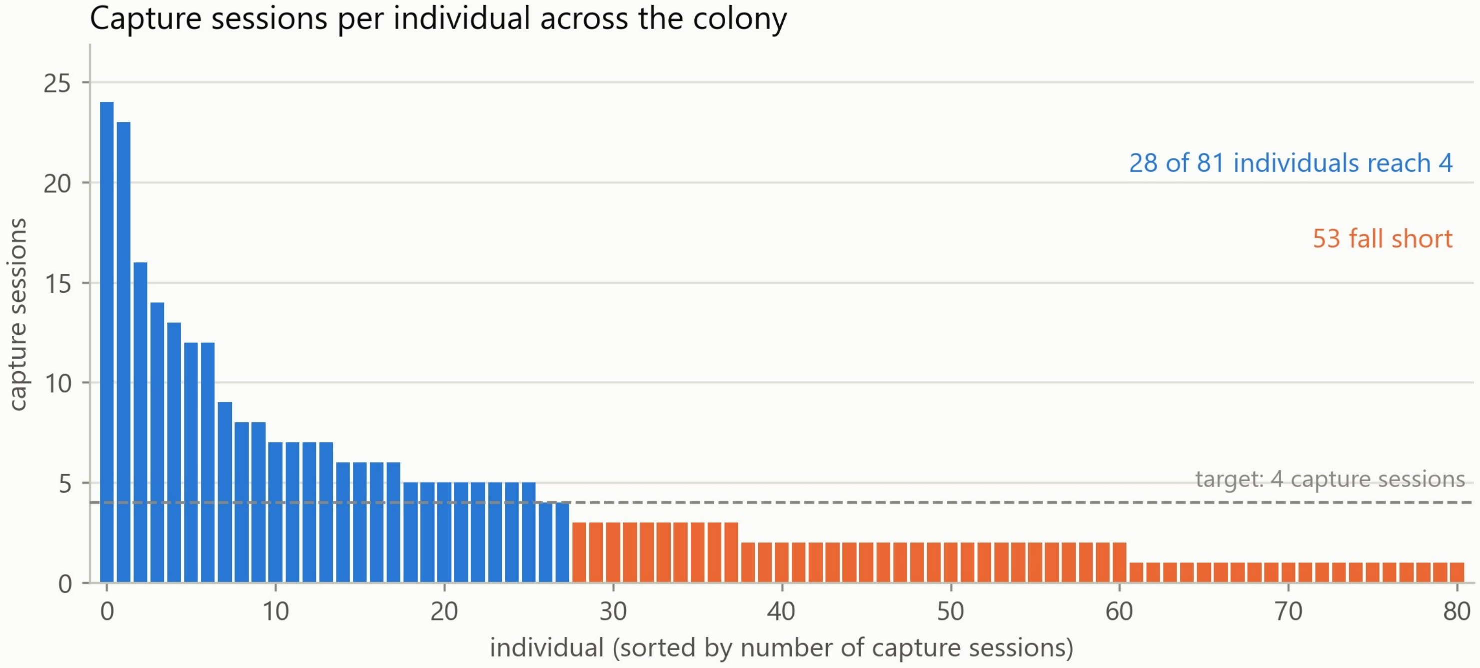
So count encounters, not files

Target four sessions per bird

Provisional — the 4–6 and 7+ groups were pooled, and sampling effort was not experimentally assigned. It is a starting point for prospective validation, not a validated threshold.

53 of 81 birds fall short

Reaching four for everyone needs 116 further individual capture encounters — roughly eight outings at about fifteen birds each, and they must fall on different days.



Vary the day, the light, the distance and the social context — and record a session ID at capture.

THE DELIVERABLE

What is deployed, and what it is for

Built and measured

The extractor retrained with the identical recipe on all 2,304 photographs of 79 individuals — no held-out set, because the estimate already exists.

A gallery enrolling all 81 colony members and all 2,307 photographs, one L2-normalised prototype apiece.

An acceptance threshold of 0.938, measured at 5% false-accept against real strangers. Smew and Skunk are enrolled without ever being trained on.

Not yet measured

The chain from an unconstrained photograph through localisation to an identity. Every number here is measured on the archive as photographed.

Rejection of blurred, rear-facing or non-penguin inputs — once all 81 are enrolled, that is what "unknown" actually means, and the threshold was never calibrated for it.

Rank-5 against the full 81-way gallery; the 35-way 0.782 cannot be assumed to transfer.

Decision support, not autonomous identification. A ranked shortlist a keeper confirms — never an identity written straight into a welfare, veterinary or breeding record.

IN CLOSING

Three things to take away

1. **In a burst-structured archive, the independent unit is the encounter — not the file.**

   Ignoring that inflated macro rank-1 by 0.531. This is not a penguin problem; it is true of any ecological image set built from bursts, repeat visits or shared conditions.

2. **Method took 0.390 to 0.559. Only a camera moves the three-session birds off 0.186.**

   Augmentation was the larger lever, ArcFace's value grows with the gallery — and neither can manufacture a photograph taken on a different day.

3. **Made to refuse real strangers, 0.559 becomes 0.123.**

   That gap, not the closed-set figure, describes deployment readiness. So it ships as a ranked suggestion for a human, and says so.

Thank you.

Aoao Guo · code and analysis: github.com/aoaoguo2003/pgs
With thanks to Robin Freeman, ZSL London Zoo, and the volunteer photographers.